Early mineral entombment as a taphonomic confound in the post-Great Oxidation Event microfossil record

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Abstract

The proposition that an apparent evolutionary radiation may reflect the opening of a preservational window is well established for the Proterozoic–Cambrian boundary, but has been less developed across the Great Oxidation Event (GOE; ∼2.4 Ga), where the diversification of morphologically diagnostic microfossils is usually read as biological. We evaluate the hypothesis that early mineral entombment, predominantly silicification with which early iron mineralisation is associated, is a taphonomic confound in the post-GOE pattern, against the published petrographic and geochemical record of seven chert-hosted biotas spanning the GOE, including the 2.45–2.21 Ga Turee Creek Group. Within the 1.88 Ga Gunflint biota, morphospecies that retain cellular contents carry 10^8–10^9 intracellular Fe atoms μm^{-3}, but this iron may reflect in vivo biomineralisation, so the observation supports the principle that fidelity requires mineralisation before decay without isolating silica, though the within-Gunflint correlation is confounded by wall-recalcitrance and the less confounded comparison is the cross-formation pattern, in which silicification varies at the facies level independently of morphotype. The hypothesis is then examined against its best available candidate, the early-silicified but contested Apex and Dresser cherts; this case cannot adjudicate the hypothesis, because their biogenicity is itself contested and is upstream of the fidelity question, but the examination exposes a deeper consequence, that silicification preserves morphology indiscriminately, entangling fidelity with biogenicity. We state explicitly the two conditions that would falsify the hypothesis. We note that the biomarker evidence for pre-GOE eukaryotes is best regarded as contaminant, while the obligate aerobicity of sterol biosynthesis gives eukaryotes a mechanistically independent reason to diversify with oxygen. The post-GOE "rise" of microfossils therefore plausibly reflects, in part, a facies-controlled preservational signal, but the partition between evolution and unmasking is unresolved and lineage-conditional. The paper advances a hypothesis; it reports no new measurements.

Keywords: Great Oxidation Event; taphonomic megabias; silicification; iron mineralisation; Archaean–Proterozoic; microfossils; molecular clocks; sterol biosynthesis.

1 Introduction

It has long been recognised that anoxia retards the degradation of sedimentary organic matter, and that the destruction of cellular morphology during early decay proceeds more slowly where molecular oxygen is scarce (Canfield, 1994; Hedges & Keil, 1995). From this premise the Archaean ocean, in which dissolved oxygen was negligible (Pavlov & Kasting, 2002) and sulfate scarce (Habicht et al., 2002; Crowe et al., 2014), might be expected to have favoured the preservation of organic-walled microfossils. The rock record appears to say the opposite: assemblages of taxonomically diagnostic microfossils become abundant and widespread only after the GOE (Javaux & Lepot, 2018; Wacey et al., 2014).

The proposition that an apparent radiation can reflect the opening of a preservational window rather than a biological event is the foundation of the taphonomic-megabias literature. Allison & Briggs (1993) first quantified the non-random, secular distribution of soft-bodied preservation through the Phanerozoic, and Butterfield (2003) argued that the distribution of Burgess-Shale-type preservation constitutes a megabias across the Cambrian explosion. Comparable arguments have been made for phosphatisation windows in the Ediacaran Doushantuo biota and for silicification in the Ediacaran (Tarhan et al., 2016). This framework is long established at the Proterozoic–Cambrian boundary; it has been less developed across the GOE.

The contribution of the present paper is therefore not the megabias concept but its specific extension to the GOE microfossil record, using the 2.45–2.21 Ga Turee Creek Group as a bridging case, and paired with a biochemical counterargument, the obligate aerobicity of sterol synthesis, that bears differentially on eukaryotes. The paper advances a hypothesis and reports no new measurements; the supporting data are semi-quantitative and drawn from a literature dominated by a few research groups, a limitation stated explicitly below.

2 Background

2.1 Redox across the GOE

The disappearance of mass-independent sulfur isotope fractionation between ∼2.45 and 2.32 Ga marks the GOE (Holland, 2006; Bekker et al., 2004; Pavlov & Kasting, 2002; Farquhar et al., 2000); the transition was protracted and regionally diachronous (Lyons et al., 2014; Philippot et al., 2018). Iron speciation indicates anoxic, ferruginous deposition throughout the Archaean and much of the Proterozoic (Canfield, 1998; Poulton & Canfield, 2005, 2011), euxinia being spatially restricted (Reinhard et al., 2009). Seawater sulfate rose across the GOE (Habicht et al., 2002; Kah et al., 2004; Crowe et al., 2014), but sulfur isotope evidence of this rise extends into the Neoarchean (Reinhard et al., 2009), and signals of oxidative weathering precede the GOE in several basins (Anbar et al., 2007).

2.2 The microfossil record and the Turee Creek Group

The Archaean record of cellular morphology is sparse and contested. The ∼3.46 Ga Apex chert filaments (Schopf, 1993) were reinterpreted as abiogenic (Brasier et al., 2002), and although better-preserved Archaean microbiotas are known from the ∼3.4 Ga Strelley Pool Formation (Sugitani et al., 2010; Delarue et al., 2020), their cellular fidelity is modest. The most directly relevant succession is the Turee Creek Group (∼2.45–2.21 Ga), which spans the GOE and preserves diverse microfossil communities in black chert (Barlow & Van Kranendonk, 2018), the taphonomy and iron mineralisation of which are documented by Lepot et al. (2017b). Gunflint-type biotas of ∼1.9 Ga age carry filaments, spheroids and spindle-shaped forms in cellular detail (Lepot et al., 2017a; Javaux & Lepot, 2018).

2.3 The silica cycle

Before the evolution of silica-biomineralising organisms the ocean was saturated with respect to amorphous silica, with dissolved concentrations of order ∼2 mM and abiological precipitation the dominant sink (Siever, 1992; Perry & Lefticariu, 2007). The capacity for early silica precipitation was therefore available on both sides of the GOE; silica was not drawn down until the much later radiation of sponges, radiolaria and diatoms.

2.4 A biochemical oxygen constraint on eukaryotes

Sterol biosynthesis is obligately aerobic: the cyclisation and oxidative modification of squalene requires molecular oxygen, on the order of eleven molecules of O_2 per sterol (Gold et al., 2017; Desmond & Gribaldo, 2009). Crown eukaryotes synthesise sterols, and the last eukaryotic common ancestor is inferred to have done so (Desmond & Gribaldo, 2009). This provides a mechanistically independent reason to expect crown-eukaryote diversification to track the availability of oxygen, though molecular-clock estimates of when that diversification occurred are themselves partly calibrated to GOE geochemistry (Gold et al., 2017), so the independence that holds is biochemical rather than chronological.

3 Material and approach

Seven chert-hosted biotas were compared (Table 1): Strelley Pool, Moodies and Tumbiana (pre-GOE), the Turee Creek Group (which spans the GOE), and the Belcher Group, Gunflint and Sokoman (post-GOE). These are the principal GOE-adjacent silicified biotas in the literature; the selection is practical rather than the product of a formal survey, and the candidate that might have served as a sharp test is examined in Section 5. The units span shallow-marine to peritidal silicified facies of comparable, low metamorphic grade. For each, published petrography, Raman spectroscopy, iron mineralogy and sulfur geochemistry were compiled; no new measurements are reported.

To reduce circularity, the degree of early silicification is assessed from criteria independent of the fidelity outcome: the timing of silica cementation relative to compaction fabrics (pre-compaction, wall-conforming microquartz or chalcedony indicating early entombment, versus late megaquartz void-fill) and the quartz cement fabric. These criteria, and the unit to which each applies, are given in Table 1.

The independence of these criteria from the fidelity outcome is logical rather than source-based. The petrographic assessments and the fidelity assessments alike are drawn from a literature dominated by a few overlapping research groups (principally Lepot, Javaux and Wacey), so the compiled comparison cannot be treated as source-independent, and the criteria-scored cross-formation plot (Fig. 2) is illustrative rather than a formal test. This limitation is restated and developed in the Discussion, where it bounds the strength of the claim.

4 Evidence

4.1 A within-assemblage covariation, and its interpretation

Within the Gunflint biota, the preservation of internal cellular structure is coupled to early mineralisation. Morphospecies retaining cellular contents carry between 3×10^8 and 10^9 intracellular Fe atoms μm^{-3}, with intra-microfossil quartz finer than the surrounding matrix; empty sheaths remain at the chert background ($\sim 1.5 \times 10^6$ atoms μm^{-3}; Fig. 1) (Lepot et al., 2017a).

This correlation carries a caveat that bears directly on the argument. Lepot et al. (2017a) interpret the intracellular iron as consistent with in vivo biomineralisation, plausibly by oxygenic photosynthesisers, rather than as a purely external precipitate. If so, the mineral phase associated with fidelity is partly a metabolic product, and the opposition between preservation and biology is less clean than a taphonomic reading would prefer. The observation nevertheless establishes the principle that fidelity required mineralisation before decay. It does not isolate silica as the agent; the assignment of silicification as the entombment medium rests on silica-specific evidence: the three-dimensional preservation of Gunflint cells within quartz, requiring silica encapsulation before compaction (Lepot et al., 2017a; Alleon et al., 2016), and wall-conforming microquartz more generally (Wacey et al., 2012).

A second, structural confound limits what the within-Gunflint correlation can establish. The morphospecies that retain cellular contents are thick-walled forms, and wall recalcitrance and the capacity to nucleate intracellular minerals are co-varying properties of the morphotype rather than independent consequences of entombment timing. The within-Gunflint covariation is therefore suggestive but cannot, by itself, distinguish entombment timing from morphotype recalcitrance as the cause. It is consequently treated here as supporting rather than flagship evidence; the less confounded comparison is the cross-formation pattern of the next subsection, where silicification varies at the facies level and is independent of morphotype (see also Discussion).

4.2 The between-assemblage pattern

The cross-formation comparison provides the more compelling illustration of the hypothesis, and for a specific reason: across the seven units, variation in silicification operates at the facies level,

Table 1: Preservation of cellular morphology in seven chert-hosted biotas, with the petrographic criterion used to assess the timing of silicification, kept separate from the fidelity outcome.

Unit (age)	GOE	Silicification criterion (independent of fidelity)	Preservation of morphology	Source
Strelley Pool (∼3.4 Ga)	pre	early silicification; wall organic matter partly replaced by microquartz/chalcedony	diverse (lenticular, spinose) but lower cellular fidelity than Gunflint	(Sugitani et al., 2010; Delarue et al., 2020; Wacey et al., 2012)
Moodies Gp (∼3.22 Ga)	pre	silicification of mat textures only; cells not encapsulated	tufted, cavity-dwelling mats; cellular detail poorly resolved	(Homann et al., 2015)
Tumbiana Fm (∼2.72 Ga)	pre	organic matter hosted in carbonate; cells not silica-encapsulated	kerogenous globules; rare filaments	(Lepot et al., 2009; Decraene et al., 2021)
Turee Creek Gp (∼2.45–2.21 Ga)	spans	early chert permineralisation with associated Fe mineralisation	diverse spheroids and filaments across the GOE	(Lepot et al., 2017b; Barlow & Van Kranendonk, 2018)
Belcher Gp (∼2.0 Ga)	post	chert replacement of stromatolitic carbonate	diverse microbiota, chert-permineralised	(Javaux & Lepot, 2018)
Gunflint Fm (∼1.88 Ga)	post	pre-compaction silica encapsulation; intra-microfossil microquartz finer than matrix	diverse, high cellular fidelity; intra-cellular Fe-silicate/carbonate	(Lepot et al., 2017a; Wacey et al., 2013; Alleon et al., 2016)
Sokoman IF (∼1.88 Ga)	post	early chert of iron-formation	Gunflint-type assemblage	(Javaux & Lepot, 2018)

across units of differing lithology and depositional setting, and is therefore independent of the morphotype-specific recalcitrance that confounds the within-Gunflint correlation. Where silicification was early and complete, cellular morphology is preserved; where it was weak or absent, only kerogen and mats survive.

Across the seven units, the preservation of cellular morphology tracks the independent indicators of early silicification rather than position relative to the GOE (Table 1; Fig. 2). The pre-GOE Strelley Pool biota preserves a diverse assemblage, albeit with lower cellular fidelity than Gunflint (Wacey et al., 2012); the pre-GOE Tumbiana and Moodies biotas preserve kerogen and mats rather than cells (Lepot et al., 2009; Homann et al., 2015); the Turee Creek Group, which spans the GOE, preserves diverse silicified microfossils with associated iron mineralisation (Lepot et al., 2017b; Barlow & Van Kranendonk, 2018), a pattern continuous across the boundary rather than stepped at it; and the post-GOE Gunflint, Sokoman and Belcher biotas are early-silicified and morphologically rich (Lepot et al., 2017a; Javaux & Lepot, 2018). Pyrite is a poor indicator of fidelity: pyritisation is patchy and destructive in Gunflint (Wacey et al., 2013; Lepot et al., 2017a) and is documented in pre-GOE Tumbiana stromatolites (Decraene et al., 2021). Plotted as a coarse fidelity-versus-silicification scatter (Fig. 2), the seven biotas fall close to a 1:1 relationship, consistent with silicification governing fidelity; the plot is criteria-scored and not source-independent, however, and is illustrative rather

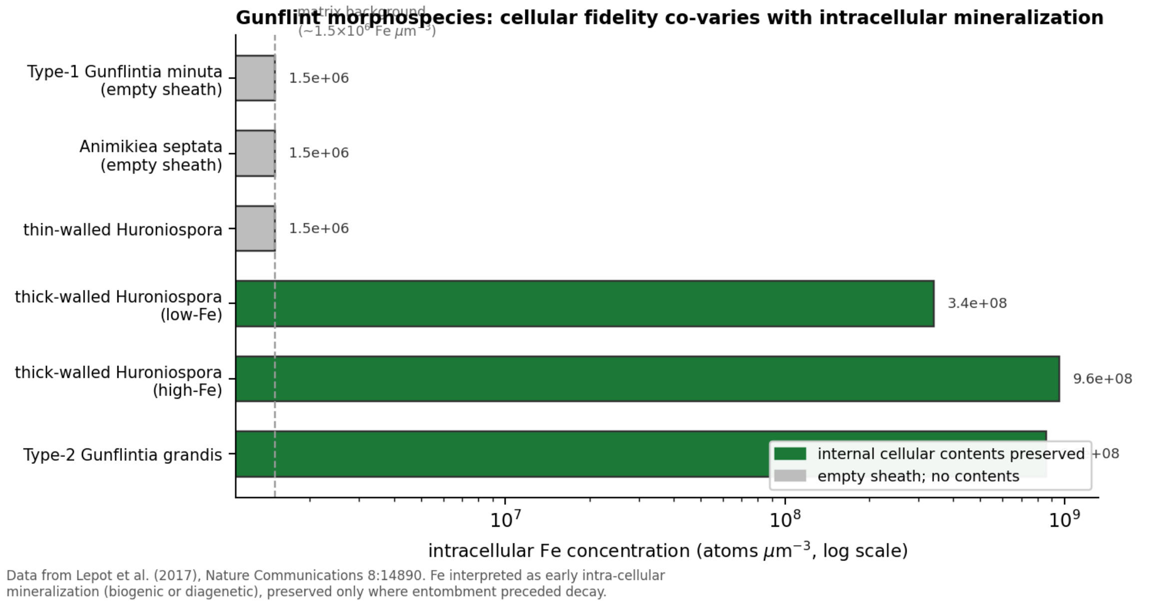

Figure 1: Intracellular Fe concentration in Gunflint morphospecies, against the preservation of cellular contents. The iron may reflect in vivo biomineralisation (Lepot et al., 2017a), so the covariation links fidelity to early mineralisation without isolating silica as the agent. Data from Lepot et al. (2017a).

than the formal regression proposed below.

4.3 Why redox and the silica cycle do not supply a GOE control

Iron speciation indicates pervasive ferruginous deposition on both sides of the GOE (Canfield, 1998; Poulton & Canfield, 2011), and the sulfur isotope signal of rising sulfate extends into the Neoarchean (Reinhard et al., 2009; Philippot et al., 2018). The sulfate rise bears on microbial sulfate reduction and the sulphurisation of organic matter, affecting the survival of bulk kerogen rather than the templating of cellular morphology. The capacity for early silica precipitation was likewise unchanging across the GOE (Siever, 1992; Perry & Lefticariu, 2007).

5 A case that cannot adjudicate: pre-GOE silicified cherts

The most promising candidate for a sharp test of the hypothesis would be a pre-GOE chert that is demonstrably early-silicified. The ∼3.46 Ga Apex chert and the ∼3.48 Ga Dresser Formation are pre-GOE, silicified, and rich in cell-like carbonaceous structures, and so might appear to provide exactly such a test. They cannot. The Apex filaments described as microfossils (Schopf, 1993) were reinterpreted as abiogenic mineral artefacts (Brasier et al., 2002), and high-resolution reinvestigation has reinforced an abiogenic, hydrothermal origin for many such structures (Wacey et al., 2014). Where biogenicity is itself contested, no observation of cellular fidelity can adjudicate the entombment-timing hypothesis, because the question of whether these structures were ever cells is logically upstream of the question of how faithfully their morphology was preserved. A chert of disputed biogenicity is therefore structurally incapable of testing the hypothesis, and we treat Apex and Dresser as informative about the nature of silicified cherts rather than as a falsification test that the hypothesis has survived.

The observation these cherts do support is narrower but worth stating plainly. Silicification preserves morphology indiscriminately — whether the substrate is biological or abiotic — so silicified cherts will contain cell-like structures of mixed origin, and the biogenicity question is upstream of the fidelity question. The scarcity of unambiguous pre-GOE microfossils is consequently not evidence that silicification was feeble in the Archaean (it demonstrably was not (Siever, 1992; Perry

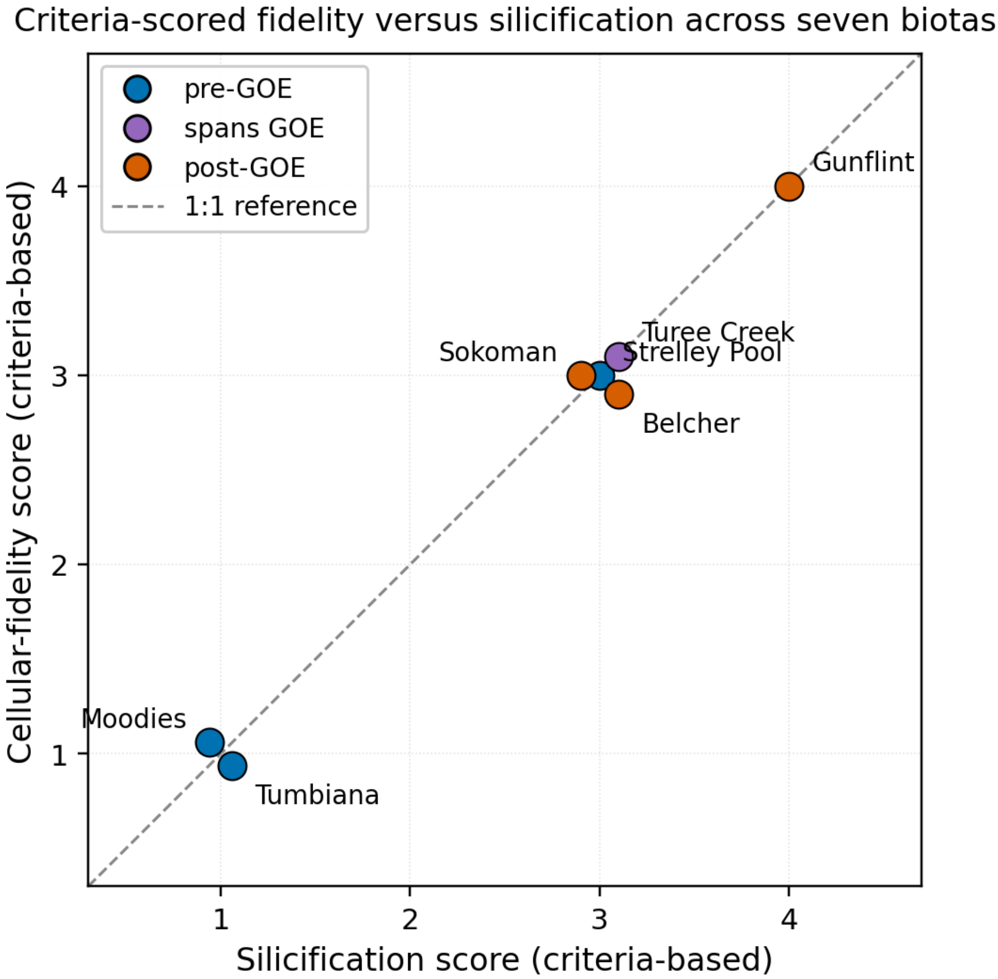

Figure 2: Criteria-scored fidelity versus silicification across seven biotas. Scores are criteria-based and compiled from the same descriptive literature; they are not source-independent (see text). The correlation is illustrative, not a formal test. Overlapping points are slightly offset for visibility. The dashed line is a 1:1 reference.

& Lefticariu, 2007)), but that silicification records both biology and abiotic look-alikes, leaving few structures that meet modern biogenicity criteria. This is a statement about the nature of silicified cherts, not a falsifier the hypothesis has passed.

What would constitute a genuine test is a chert carrying microfossils of undisputed biogenicity, with demonstrably early silicification, and varying cellular fidelity. A single such unit in which early entombment coexisted with poor fidelity would falsify the hypothesis outright. No such case is presently known: every well-preserved early-silicified biota of accepted biogenicity also preserves morphology well, and every pre-GOE chert of disputed biogenicity cannot serve as a test for the reason given above.

6 Origins, first appearances, and the sterol constraint

Molecular clocks place the stem origin of cyanobacteria, and oxygenic photosynthesis, in the Archaean (Fournier et al., 2021), whereas the crown-group diversification of extant Oxyphotobacteria postdates the rise of oxygen (Shih et al., 2017). Molecular clock estimates for the last eukaryotic common ancestor are model-dependent but several precede the first widely accepted eukaryotic microfossils (Betts et al., 2018; Eme et al., 2014) (Fig. 3). The biomarker record cannot resolve the resulting gap: the steranes and 2-methylhopanes reported from ~2.7 Ga shales (Brocks et al., 1999) reside in

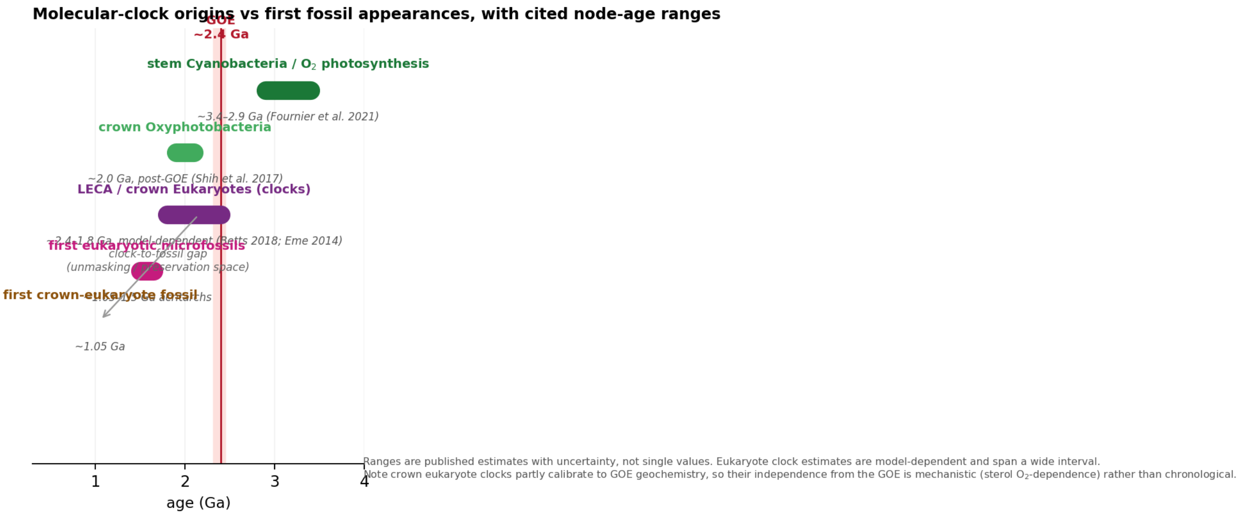

Figure 3: Molecular clock estimates of lineage origin against first fossil appearances, with published node-age ranges. The stem origin of cyanobacteria lies in the Archaean (Fournier et al., 2021); crown Oxyphotobacteria diversify around or after the GOE (Shih et al., 2017); eukaryote estimates are model-dependent (Betts et al., 2018; Eme et al., 2014) and partly calibrated to GOE geochemistry.

younger carbonate veins (Rasmussen et al., 2008) and are indistinguishable from procedural blanks in ultraclean cores (French et al., 2015).

The obligate aerobicity of sterol synthesis (Gold et al., 2017; Desmond & Gribaldo, 2009) cuts across a purely preservational reading for eukaryotes: crown eukaryotes have a mechanistic reason to diversify once oxygen became available, regardless of preservation. The clock-to-fossil gap for eukaryotes is thus not necessarily a preservational lag and may reflect genuine, oxygen-limited diversification. The unmasking hypothesis is consequently better supported for cyanobacteria, whose metabolism clocks to the Archaean, than for sterol-synthesising eukaryotes, and any reading must be lineage-conditional.

7 Discussion

The compiled record is consistent with the hypothesis that early mineral entombment is a significant confound in the post-GOE microfossil record. The defensible position is narrow. The preservation of cellular morphology in Precambrian cherts is conditioned by early mineralisation, predominantly silicification, that operated throughout but was expressed unevenly, depending on depositional facies and the timing of cementation relative to decay (Fig. 4). Where entombment was early and complete, morphology survives; where it was weak, kerogen and mats remain. Oxygenation plausibly modified organic-matter decay and kerogen sulphurisation (Lepot et al., 2009; Decraene et al., 2021) but did not gate the silicification on which cellular fidelity depends.

The limitations of these compiled observations must be stated plainly, since they define the status of the contribution. The quantitative correlation rests on a single locality (Gunflint), a single element (iron), and a single study in which both the mineralisation and the fidelity were documented by the same authors (Lepot et al., 2017a); the within-assemblage covariation is therefore not source-independent. A structural confound compounds this: within Gunflint the morphospecies retaining contents are thick-walled forms, and wall recalcitrance and mineral-nucleation potential are co-varying properties of the morphotype rather than independent consequences of entombment timing. This does not rescue a purely biological reading (morphotype-selective preservation is itself a taphonomic bias) but it means the within-Gunflint evidence cannot, by itself, isolate entombment timing as the cause. The cross-formation comparison, which uses criteria independent of the fidelity outcome and varies silicification at the facies level, is the more robust comparison precisely because

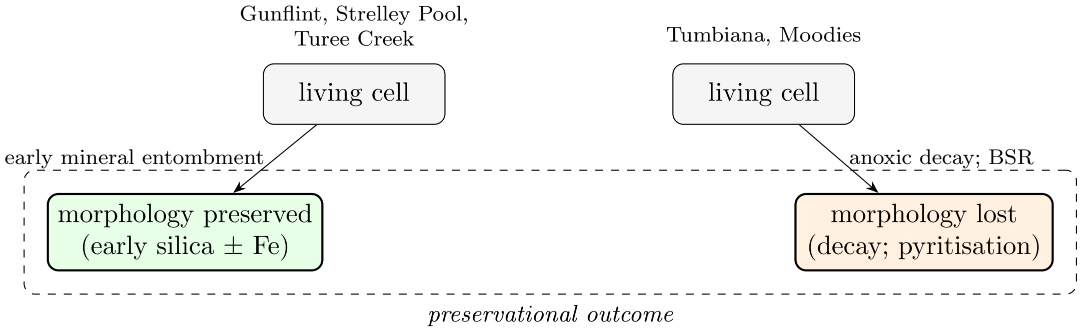

Figure 4: Preservational pathways. Where early mineral entombment outpaces decay, morphology is retained; where decay precedes mineralisation, morphology is lost and kerogen survives.

it is less susceptible to this confound; but its criteria and fidelity assessments are likewise drawn from a literature dominated by a few overlapping research groups (principally Lepot, Javaux and Wacey), so the independence is logical rather than source-based.

The hypothesis would be falsified by (i) a pre-GOE chert bearing confirmed genuine microfossils with demonstrably early silicification and poor cellular fidelity, or (ii) a within-formation regression showing no relationship between fidelity and a quantitative silicification index scored blind and across research groups. Neither is known; both are testable, and the second is achievable with existing material. The programme needed to pursue them is specified below.

The implications are qualified accordingly. Abundance and diversity counts across the GOE cannot be read as proxies for biomass without facies correction. The clock-to-fossil gap is consistent with a preservational lag but, for sterol-synthesising lineages, may equally reflect oxygen-limited diversification. For the search for ancient biosignatures, the corollary is that the preservation of morphology requires a silica-precipitating facies rather than the presence of oxygen.

7.1 What would be needed: an experimental and analytical programme

The hypothesis is testable; we specify three approaches in order of decisiveness.

(a) Controlled silicification–decay experiments. The confound that most limits the within-Gunflint evidence is the co-variation of wall recalcitrance and mineral-nucleation potential with morphotype. A laboratory programme can hold morphotype constant and vary only the timing of silica addition. Microbial cultures spanning thick- and thin-walled morphotypes (e.g. cyanobacteria and other bacteria) would be incubated under factorial O_2 × sulfate × Fe conditions, with controlled silica added at defined intervals after death. Blind-scored cellular fidelity, silica nucleation timing, and intracellular mineralisation would then reveal whether entombment timing — rather than wall recalcitrance — is the causal variable. Existing experimental work shows that early entombment within silica minimises the molecular degradation of microorganisms during advanced diagenesis (Alleon et al., 2016), providing a mechanistic basis for the timing effect the hypothesis predicts.

(b) A blind, cross-group within-formation regression. This is the test the present paper proposes but does not perform. Cellular fidelity and a quantitative silicification index would be scored independently by different research groups, blind to one another's scores, on 30–50 microfossil populations per formation, and regressed while controlling for thermal maturity (Raman R_2). A null relationship across formations of comparable grade would weigh against the hypothesis; a consistent positive relationship would support it.

(c) A systematic survey for the falsifier. A single pre-GOE chert (or indeed any-age chert) carrying microfossils of undisputed biogenicity, demonstrably early silicification, and poor cellular fidelity would falsify the hypothesis outright. A targeted search of curated chert collections, screened first for biogenicity and silicification timing and then scored blind for fidelity, is the most economical route to a decisive test. No such case is currently known.

8 Conclusions

We advance the hypothesis that early mineral entombment, predominantly silicification, is a taphonomic confound in the apparent post-GOE diversification of microfossils, situating the argument within the established taphonomic-megabias tradition and extending it to a boundary at which it has been less developed. The published record is consistent with the hypothesis, but does not sustain the stronger claim that oxygenation governed preservation: the within-Gunflint correlation is single-locality, single-element, possibly biogenic, and confounded by wall recalcitrance; the cross-formation pattern, which is less susceptible to that confound, provides the more robust comparison but remains semi-quantitative and not source-independent; and the best available candidate for a sharp test, the early-silicified Apex and Dresser cherts, cannot adjudicate the hypothesis because their biogenicity is itself contested and upstream of the fidelity question. The sterol-biochemical constraint and the contaminated biomarker record leave the partition between evolution and unmasking unresolved and lineage-conditional. We state the two conditions that would falsify the hypothesis and specify the experimental and analytical programme needed to test them, one element of which is achievable on existing material.

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